📘 RELATIONS & FUNCTIONS – SHORT NOTES (Class 12)

🔷 1. Relation kya hota hai?

Agar A aur B do sets hain, to

A × B ka koi bhi subset = Relation

👉 Example:

A = {1,2}, B = {3,4}

A × B = {(1,3),(1,4),(2,3),(2,4)}

🔷 2. Types of Relations

✅ (i) Reflexive Relation

Agar har element ke liye (a,a) present ho

👉 $(a,a) \in R$

✅ (ii) Symmetric Relation

Agar $(a,b) \in R \Rightarrow (b,a) \in R$

✅ (iii) Transitive Relation

Agar (a,b) & $(b,c) \in R \Rightarrow (a,c) \in R$

✅ (iv) Equivalence Relation

Jo Reflexive + Symmetric + Transitive ho

🔷 3. Function kya hota hai?

Relation ka special case:

👉 Har input ka exactly 1 output hona chahiye

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🔷 4. Types of Functions

✅ (i) One-One (Injective)

Different inputs → different outputs

✅ (ii) Onto (Surjective)

Har element of codomain covered ho

✅ (iii) Bijective

One-One + Onto

👉 Inverse exist karta hai

✅ (iv) Many-One

Different inputs → same output allowed

🔷 5. Important Concepts

👉 Domain

Allowed inputs

👉 Codomain

Possible outputs

👉 Range

Actual outputs

🔷 6. Algebra of Functions

Let f(x), g(x)

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$(f + g)(x) = f(x) + g(x)$

$(f - g)(x) = f(x) - g(x)$

$(fg)(x) = f(x) \cdot g(x)$

$(f/g)(x) = f(x)/g(x), g(x) \neq 0$

🔷 7. Composition of Functions

$(f \circ g)(x) = f(g(x))$

🔷 8. Inverse Function

👉 Sirf bijective function ka inverse hota hai

$f^{-1}(x)$ nikalne ke steps:

$y = f(x)$

x aur y swap karo

y ko solve karo

🔷 9. Important Points (Exam Tips 🔥)

Function me 1 input → 2 output ❌ allowed nahi

Vertical line test → function check

Bijective ⇒ inverse possible

Equivalence relation ⇒ classes bana sakte hain

Good luck friends

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